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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

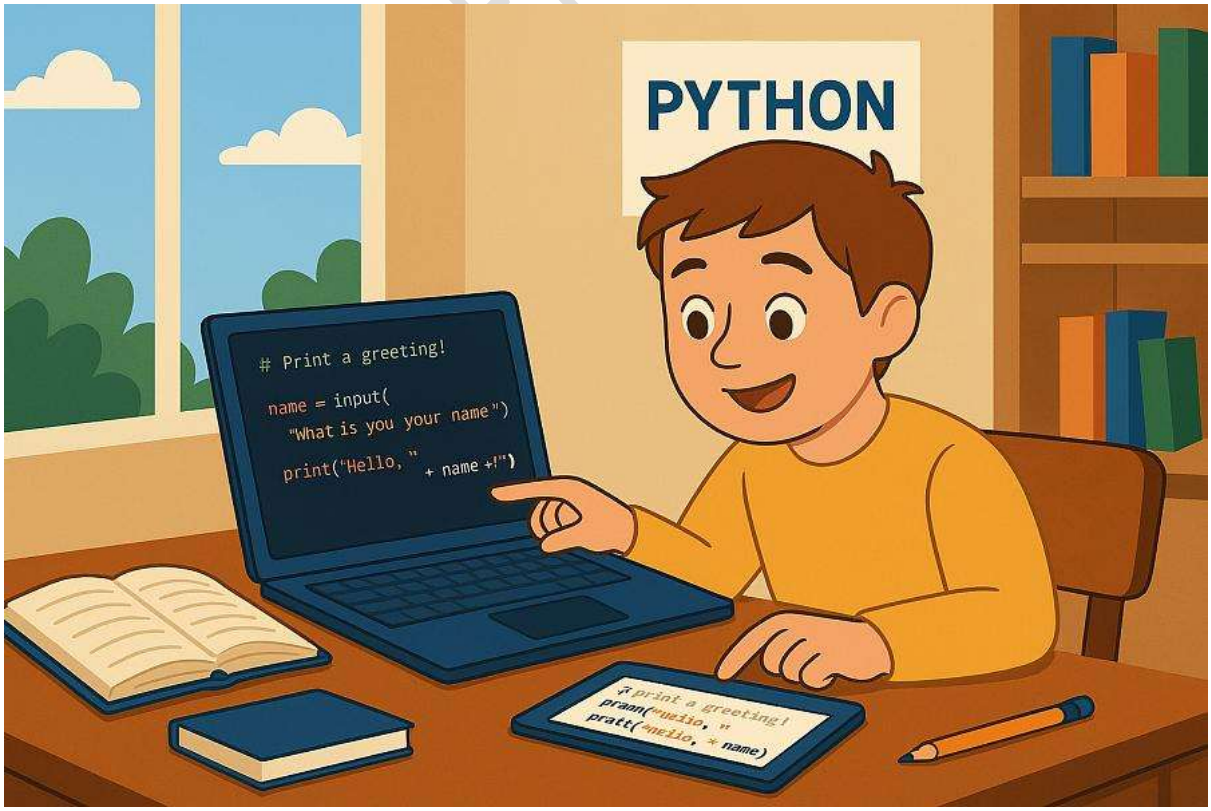
Daily Teaching : 2 Hours

Total Classes : 100

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Level 2 – Young Technologist (7th–8th Standard)

Lesson 17: Input & Output (I/O) in Python



Learning Objectives

After completing this lesson, students will be able to:

- ✅ Understand what Input and Output are.
- ✅ Display text using print().
- ✅ Take user input using input().
- ✅ Store user input in variables.
- ✅ Create simple interactive Python programs.
- ✅ Differentiate between input and output.

What is Input?

Input means giving information to the computer.

The computer waits for the user to enter data.

Examples:

- Typing your name
- Entering your age
- Typing marks
- Entering your favorite color

Real-life Example

Teacher asks:

"What is your name?"

Student replies:

"Rahul"

Here,

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Teacher asks → **Input**

Student answers → **Output**

What is Output?

Output is the result shown by the computer.

Examples:

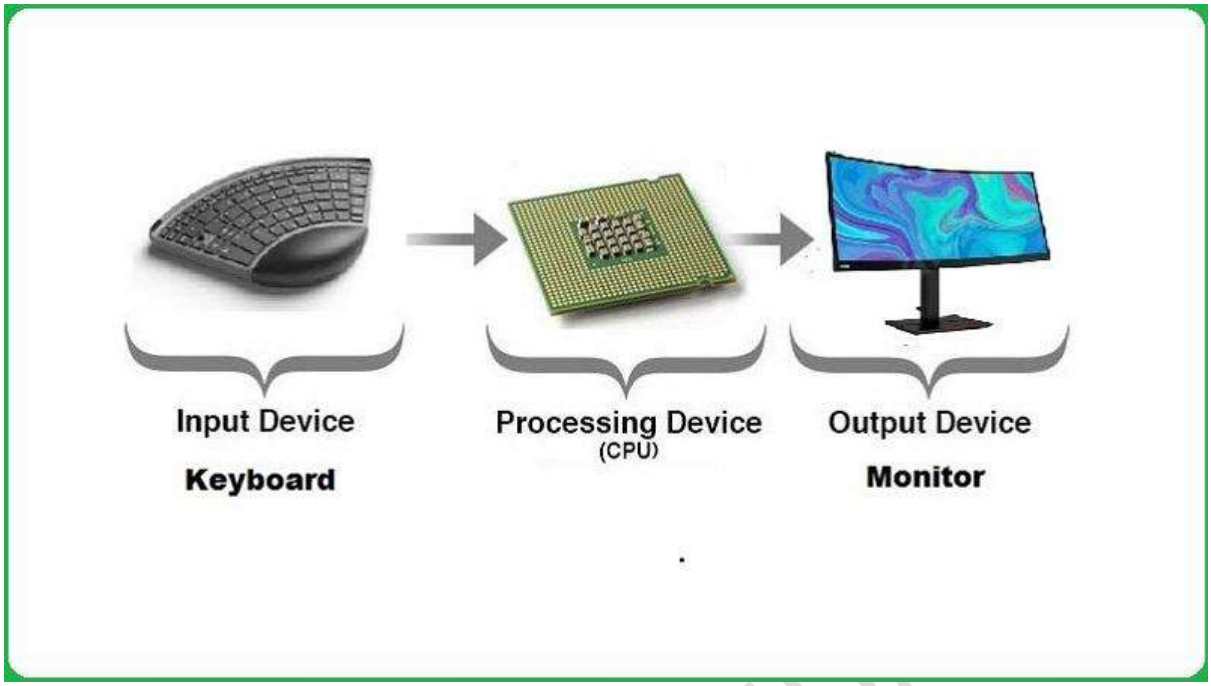
- Displaying text
- Showing marks
- Printing a message
- Displaying calculations

Daily Life Example

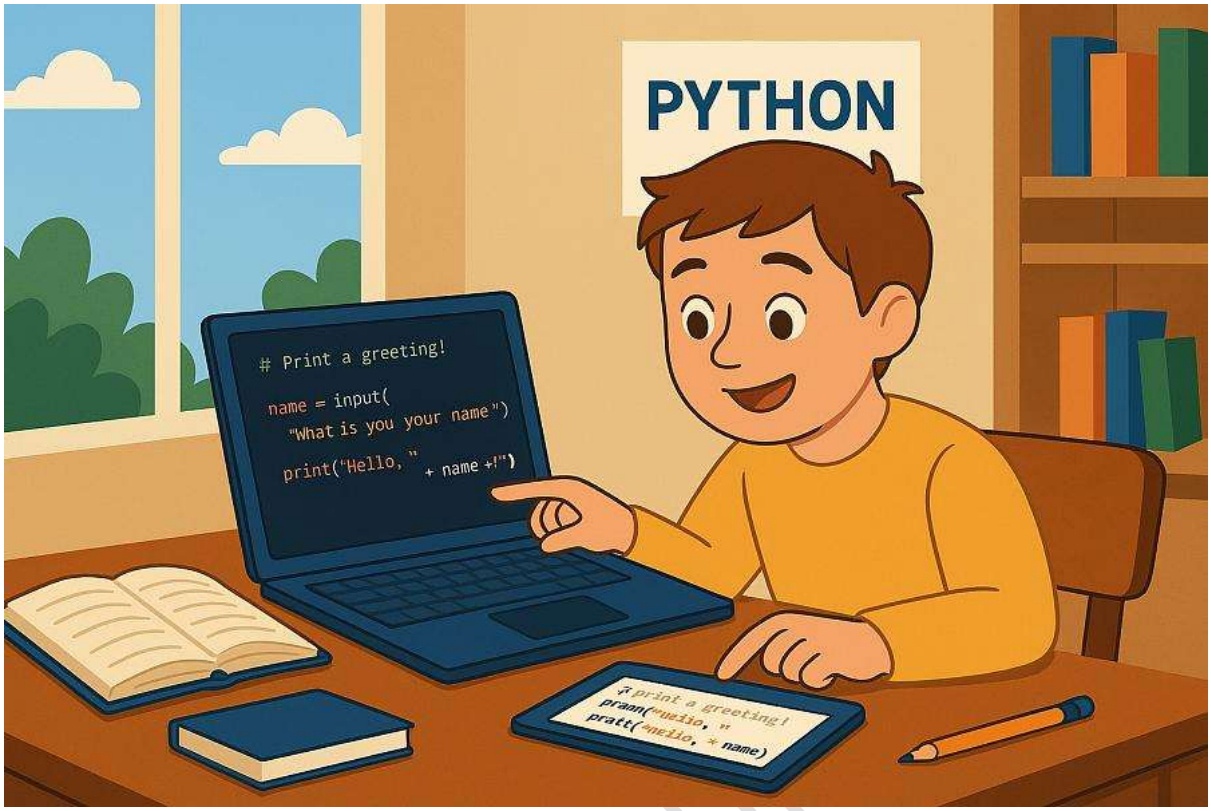
Situation	Input	Output
ATM	PIN Number	Cash
Mobile	Password	Home Screen
Calculator	Numbers	Answer
Computer	Keyboard	Screen Display

Input and Output Process

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Input Device
Keyboard
Processing Device
(CPU)
Output Device
Monitor





Flow

Student

↓

Keyboard

↓

Computer

↓

Monitor

Input → Processing → Output

Python Output

Python uses

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print()

to display output.

Example

print("Hello Students")

Output

Hello Students

Example

print("Welcome to Python")

Output

Welcome to Python

Printing Numbers

print(10)

Output

10

Printing Multiple Values

print("Age:",12)

Output

Age: 12

Printing Variables

name="Rahul"

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print(name)

Output

Rahul

Python Input

Python uses

input()

to get information from the user.

Example

name=input()

The computer waits for the user to type.

Input Example

name=input("Enter your name: ")

print(name)

Example

Enter your name:

Rahul

Rahul

How Input Works

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Computer:

Enter your name:

↓

Student types

Rahul

↓

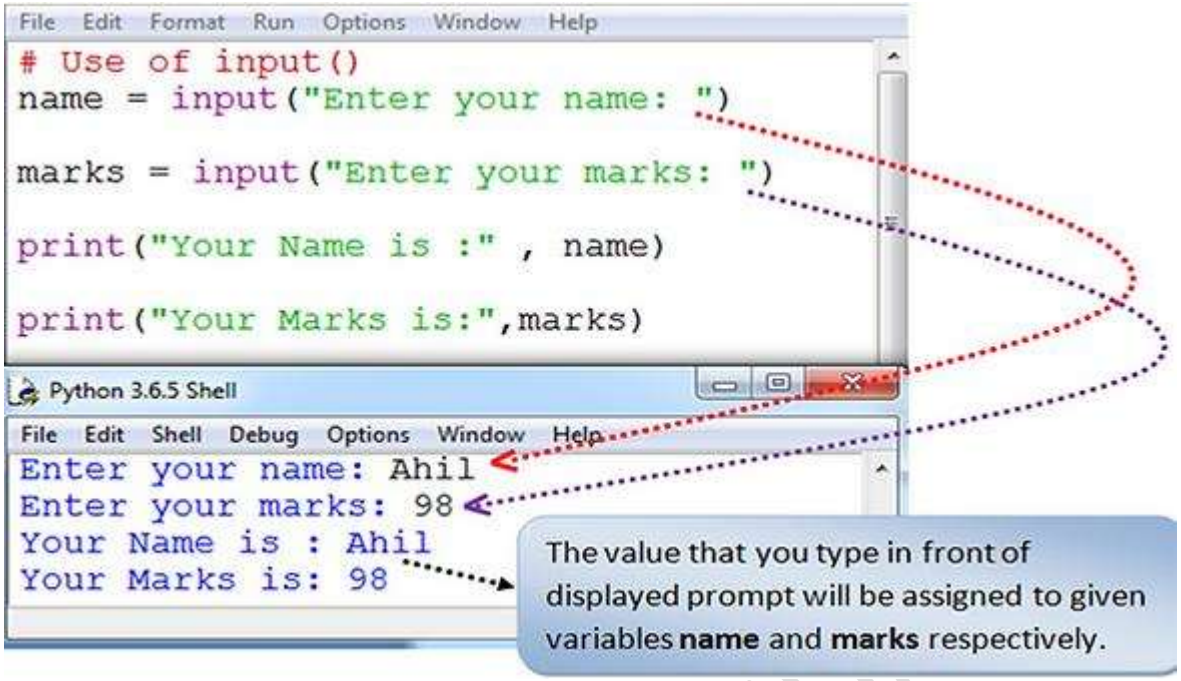
Python stores it

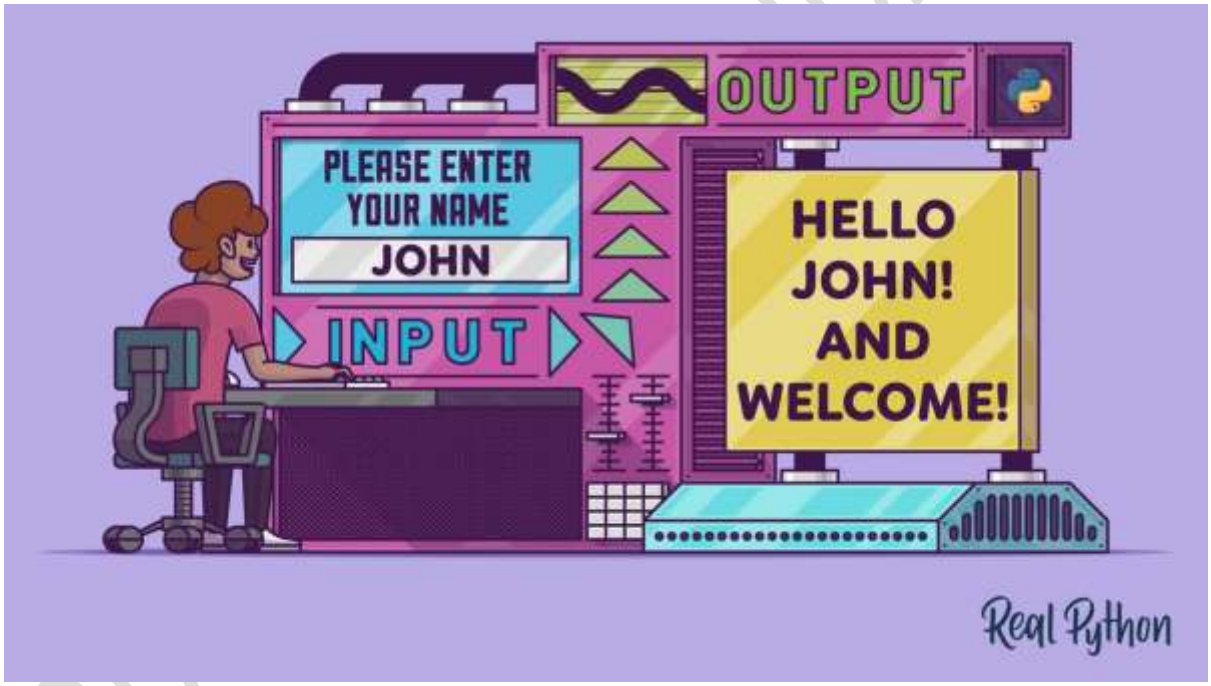
↓

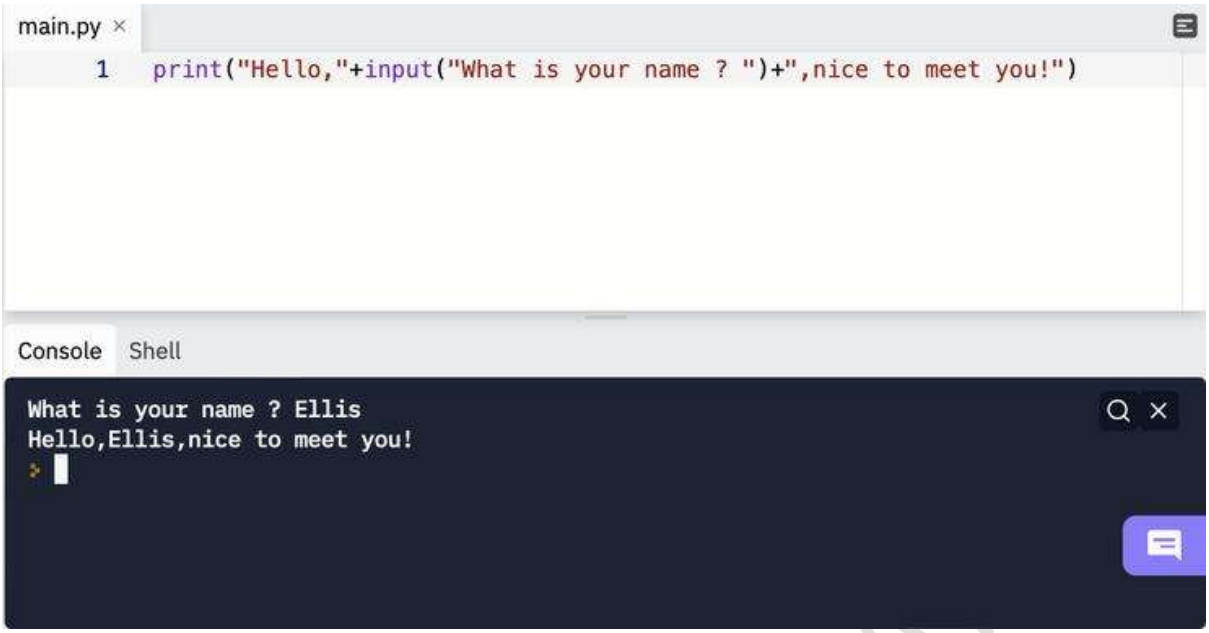
Prints Rahul

User Input Example

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File Edit Format Run Options Window Help
Use of input()
name = input("Enter your name: ")
marks = input("Enter your marks: ")
print("Your Name is :" , name)
print("Your Marks is:",marks)
Python 3.6.5 Shell
File Edit Shell Debug Options Window Help
Enter your name: Ahil
Enter your marks: 98
Your Name is : Ahil
Your Marks is: 98
The value that you type in front of displayed prompt will be assigned to given variables name and marks respectively.


OUTPUT
PLEASE ENTER YOUR NAME
JOHN
INPUT
HELLO JOHN! AND WELCOME!
Real Python



Using Variables with Input

name=input("Enter Name: ")

print("Hello",name)

Output

Enter Name:

Rahul

Hello Rahul

Example 2

city=input("Enter City: ")

print("Welcome to",city)

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Output

Enter City:

Hyderabad

Welcome to Hyderabad

Example 3

```
school=input("School Name: ")

print("School:",school)
```

Example 4

```
food=input("Favorite Food: ")

print(food)
```

Example 5

```
animal=input("Favorite Animal: ")

print("You like",animal)
```

Input Always Returns Text

Important:

input() always gives **text (string)**.

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Example

```
age=input("Enter Age:")


print(age)
```

If user types

12

Python stores

"12"

Converting Input into Numbers

Use

int()

Example

```
age=int(input("Enter Age: "))


print(age)
```

Now age becomes a number.

Example

```
num1=int(input("Enter First Number: "))


num2=int(input("Enter Second Number: "))


print(num1+num2)
```

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Input

10

20

Output

30

Without int()

```
num1=input()

num2=input()

print(num1+num2)
```

Input

10

20

Output

1020

Because both are text.

With int()

```
num1=int(input())

num2=int(input())
```

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print(num1+num2)

Output

30

Activity 1

Ask the student's name.

name=input("Enter Name:")

print("Hello",name)

Activity 2

Ask favorite game.

game=input("Favorite Game:")

print(game)

Activity 3

Ask school name.

school=input("School:")

print(school)

Activity 4

Ask age.

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```
age=int(input("Age:"))

print(age)
```

Activity 5

Add Two Numbers

```
a=int(input("First Number:"))

b=int(input("Second Number:"))

print(a+b)
```

Activity 6

Multiply Two Numbers

```
a=int(input())

b=int(input())

print(a*b)
```

Activity 7

Greeting Program

```
name=input("Name:")
```

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```
print("Good Morning",name)
```

Activity 8

Student Details

```
name=input("Name:")

age=input("Age:")

city=input("City:")

print(name)
print(age)
print(city)
```

Practice Programs

Program 1

```
print("Welcome")
```

Program 2

```
name=input("Name:")

print(name)
```

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Program 3

```
color=input("Favorite Color:")


print(color)
```

Program 4

```
fruit=input("Favorite Fruit:")


print(fruit)
```

Program 5

```
num=int(input())


print(num)
```

Practical Lab Activities

Practical 1

Print your name five times.

Practical 2

Ask your friend's name.

Practical 3

Display your school name.

Practical 4

Add two numbers.

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Practical 5

Multiply two numbers.

Practical 6

Create a greeting program.

Practical 7

Create a simple calculator (addition only).

Practical 8

Display student details.

Classroom Activity

Pair Activity

Student A enters:

Name

Age

City

Student B's program displays:

Welcome Rahul

Age 12

City Hyderabad

Fun Challenge

Create a program that asks:

- Name
- Favorite Animal

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- Favorite Color
- Favorite Subject

Then prints:

Hello Rahul

Your Favorite Animal is Dog

Your Favorite Color is Blue

Your Favorite Subject is Science

Common Mistakes

❌ Forgetting quotation marks

```
print(Hello)
```

✅ Correct

```
print("Hello")
```

❌ Forgetting brackets

```
print "Hello"
```

✅ Correct

```
print("Hello")
```

❌ Adding text instead of numbers

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"10"+"20"

Output

1020

Use

int()

Keywords

- **Input** – Information entered by the user.
- **Output** – Information shown by the computer.
- **print()** – Displays output.
- **input()** – Reads data from the user.
- **Variable** – Stores information.
- **String** – Text data.
- **Integer (int)** – Whole number data.

Lesson Summary

- Input means entering data into the computer.
- Output means displaying information from the computer.
- print() shows messages or values.
- input() takes information from the user.
- Variables store the entered data.
- Use int() to convert numeric input into numbers.
- Interactive programs make Python fun and useful.

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Quiz

Multiple Choice Questions

1. Which function displays output?

A) input()

B) print() ✅

C) int()

D) str()

2. Which function takes user input?

A) print()

B) input() ✅

C) type()

D) len()

3. Input from input() is stored as:

A) Integer

B) Float

C) String ✅

D) Boolean

4. Which function converts text into an integer?

A) float()

B) input()

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C) print()

D) int() ✅

5. What does Output mean?

A) Entering data

B) Displaying results ✅

C) Deleting files

D) Saving programs

Homework

1. Write a program to ask your name and greet the user.
2. Write a program to ask your age and display it.
3. Write a program to ask your favorite fruit and color.
4. Write a program to add two numbers entered by the user.
5. Write a program to display your school name and city.

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